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Heat exchanger including cross channel communication

Abstract

In some examples, a heat exchanger includes a first flow channel having a first flow channel inlet and a first flow channel outlet. The heat exchanger also includes a second flow channel having a second flow channel inlet and a second flow channel outlet. A fin may separate the first flow channel from the second flow channel. The fin may define at least one aperture configured to allow fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of foreign object debris.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates to heat exchangers.

BACKGROUND

(2) Vehicles may include various systems that work together to provide power to the vehicle. For example, a vehicle such as an aircraft may include a gas turbine engine and systems that support the gas turbine engine in the generation of power to power an aircraft. Due to the operating systems associated with the vehicle, one or more heat exchanger may be employed to control a temperature of various fluids associated with the vehicle. In certain instances, due to the location of the heat exchanger on the vehicle, a size of the heat exchanger may be restricted to fit within a given space onboard the vehicle. Furthermore, the circumstances may sometimes dictate that the heat exchanger be located in a place on board the vehicle that is subject to conditions including elevated temperatures, pressures, or external conditions including dust, sand, ice, hail, or the like. The heat exchanger may be manufactured to function in such a hostile environment.

SUMMARY

(3) In some examples, the disclosure relates to a heat exchanger which includes adjacent flow channels separated by a fin, and a method of manufacturing a heat exchanger. The heat exchanger

may include a fin which defines at least one aperture configured to allow fluid to flow between a first flow channel and a neighboring second flow channel. In operation, if an inlet of the first flow channel or an inlet of the second flow channel becomes constricted, such as when the heat exchanger is included on a rotary wing aircraft (e.g., a helicopter) or another vehicle that may operate in a sandy or dusty environment where foreign object debris (FOD) may be ingested by the heat exchanger, the flow through said flow channel will be reduced. Inclusion of a heat exchanger fin or fins which define an aperture or apertures that allow for fluid to flow between channels may mitigate or eliminate efficiency losses due to FOD constriction or blockage, without significant increases in size or weight of the heat exchanger. In fact, the resulting heat exchanger may advantageously be lighter than a similar heat exchanger that does not include apertures configured to allow for cross channel communication.

(4) In some examples, the disclosure relates to a heat exchanger which includes a first flow channel having a first flow channel inlet and a first flow channel outlet. The heat exchanger also includes a second flow channel having a second flow channel inlet and a second flow channel outlet. A fin separates the first flow channel from the second flow channel. The fin defines at least one aperture configured to allow fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of one or more of sand, foreign object debris (FOD), or other contaminants.

(5) In some examples, the disclosure is directed to a technique for manufacturing a heat exchanger. The technique includes forming a first flow channel having a first flow channel inlet and a first flow channel outlet. The technique also includes forming a second flow channel, the second flow channel having a first flow channel inlet and a first flow channel outlet. The technique includes separating the first flow channel from the second flow channel with a fin. The fin defines at least one aperture which allows the fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of one or more of sand, foreign object debris (FOD), or other contaminants.

(6) This summary is intended to provide an overview of the subject matter described in this disclosure. It is not intended to provide an exclusive or exhaustive explanation of the systems, devices, and methods described in detail within the accompanying drawings and description below. Further details of one or more examples are set forth in the accompanying drawings and the description below. Other features, objects, and advantages will be apparent from the description and drawings, and from the statements provided below.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) FIG. 1 is a functional block diagram of a thermodynamic system, which, in some examples, is onboard a vehicle and includes an exemplary heat exchanger according to the present disclosure.

(2) FIG. 2 is a conceptual diagram illustrating an example plate fin heat exchanger core with one side for cold flow and the other side for hot flow.

(3) FIGS. 3A-3B illustrate a portion of an example heat exchanger which includes a fin that defines one or more apertures configured to allow for cross-channel communication, according to some examples of the present disclosure.

(4) FIGS. 4A-4B illustrate a portion of an example heat exchanger which includes a wavy fin that defines one or more apertures configured to allow for cross-channel communication, according to some examples of the present disclosure.

(5) FIG. 5 illustrates an example fin of an example heat exchanger, according to some examples of the present disclosure.

(6) FIG. 6 is a flowchart illustrating an example technique for manufacturing a heat exchanger.

DETAILED DESCRIPTION

(7) In some examples, the disclosure describes heat exchangers and techniques for making such heat exchangers. Heat exchangers are employed in a variety of applications, including vehicles (e.g., aircraft such as helicopters) which operate in extreme environments at extreme temperatures. The disclosed heat exchangers may also be used in environmental control systems (ECS), or in stationary applications operating in extreme environments. Due to the operating conditions associated with the vehicle, the size of the heat exchanger may be restricted to fit within a given space onboard the vehicle. The restriction in size of the heat exchanger may result in limited heat transfer. Accordingly, it is desirable to provide a heat exchanger that is sized to fit within a restricted space onboard a vehicle, while providing the required heat transfer that meets the needs of a particular working fluid. In addition, it is desirable to minimize a weight of a heat exchanger or heat exchangers of a particular vehicle.

(8) Some extreme environments mentioned above include environments where foreign object debris (FOD) may be ingested by the heat exchanger. FOD generally refers to any solid material or particle in the fluid stream entering the heat exchanger. For example, FOD may include, but is not limited to, sand, dust, pebbles, ice, hail, other contaminants, combinations thereof, or the like. There is increasing demand for vehicles, such as aircraft (e.g., rotary wing aircraft such as a helicopter), to operate in such environments. Therefore, there is a need to provide a heat exchanger which mitigates or eliminates the deleterious effects of FOD, which may include reduced efficiency of the heat exchanger due to constriction or plugging of heat exchanger flow channels.

(9) One way to address problems associated with FOD contamination without increasing size of the heat exchanger is to make a flow channel inlet and the associated flow channel larger in cross-sectional area, such that FOD (e.g., a pebble) may more easily pass through the device without becoming hung up and constricting the fluid path through the flow channel. However, increasing the cross-section of the flow channel may reduce efficiency of the heat exchanger by reducing available surface area for heat transfer, so increasing the cross-section of the heat exchanger may not be desirable or possible within the limits of certain applications.

(10) Heat exchangers according to the present disclosure may address problems associated with FOD contamination by including one or more apertures in a fin separating flow channels, promoting cross channel fluid communication. As such, heat exchangers according to the present disclosure may deliver improved heat exchanger efficiency while operating in an environment where FOD ingestion may occur, while mitigating or eliminating the overall heat transfer efficiency losses due to reduced effective heat transfer area.

(11) With reference to FIG. 1, a functional block diagram of thermodynamic system **10** onboard vehicle **12**, which may be an aircraft such as a helicopter, is shown. In this example, thermodynamic system **10** includes heat exchanger **14**, which transfers heat from hot working fluid **16** to cold working fluid **18** to reduce a temperature of hot working fluid **16**. Hot working fluid **16** is provided from hot fluid source **20**, while cold working fluid **18** is provided from cold fluid source **23**. In some examples, hot fluid source **20** includes, but is not limited to: a compressor section associated with the gas turbine engine that supplies compressed air as hot working fluid **16**; a lubrication system that supplies engine lubrication fluid, such as oil, as hot working fluid **16**; an electronic system that supplies coolant as hot working fluid **16**; etc. In some examples, cold fluid source **23** may include, but is not limited to: a fan of a gas turbine engine associated with vehicle **12** that supplies fan air as cold working fluid **18**; a ram air inlet associated with vehicle **12** that supplies atmospheric air as cold working fluid **18**; a refrigerated coolant supply associated with vehicle **12** that supplies refrigerated coolant as cold working fluid **18**; etc. It should be noted that although described herein as a single heat exchanger **14** within a single thermal management system **10** onboard vehicle **12**, in some examples, thermal management system **10** may include more than one heat exchanger **14** according to the present disclosure, and/or vehicle **12** may include more than one thermal management system **10** which includes heat exchanger **14**.

(12) As will be discussed, heat exchanger **14** receives hot working fluid **16** and cold working fluid **18**, and via convective and conductive heat transfer, transfers heat from hot working fluid **16** to cold working fluid **18** to output reduced temperature working fluid **16'** and increased temperature working fluid **18'**. Thus, in some examples, heat exchanger **14** includes core **15** fluidically coupled to (i.e., in fluid communication with) hot manifold or inlet **30**, cold manifold or inlet **32**, reduced temperature hot manifold or first outlet **34**, and increased temperature cold manifold or second outlet **36**. As will also be discussed, given the structure of core **15** of heat exchanger **14**, heat exchanger **14** may be manufactured to be positioned within limited or restricted spaces, while still providing sufficient surface area for convective heat transfer to meet the cooling requirements associated with the thermodynamic system **10** with a reduced weight. Core **15** may also provide features discussed elsewhere herein which mitigate problems associated with FOD. For example, core **15** includes a fin (not illustrated in FIG. **1**) which separates adjacent flow channels within core **15**. The fin defines at least one aperture configured to allow for fluidic communication of hot working fluid **16** and/or cold working fluid **18** in between an inlet of the flow channels and an outlet of the flow channels, so that if one of the flow channel inlets becomes constricted (e.g., blocked by FOD), the constricted flow channel may receive working fluid from the adjacent (less constricted) flow channel, and the working surface area associated with the constricted flow channel may continue to work (e.g., transfer heat from hot working fluid **16** to cold working fluid **18** during operation).

(13) Generally, in the example of compressor air as hot working fluid **16**, the compressor air has a temperature, which may be in excess of 600 to 1200 degrees Fahrenheit, while cold working fluid **18**, for example, cooling air, has a temperature of about -50 degrees Fahrenheit to 300 degrees Fahrenheit. In this example, core **15** promotes heat transfer between hot working fluid **16** and cold working fluid **18** at a rate of about 100 to about 1000 British Thermal Units per minute (BTU/min). It should be noted that the above is merely an example, and that the values may vary based on the configuration, size and requirements of core **15** of heat exchanger **14**.

(14) In some examples, core **15** of heat exchanger **14** may include a metal or metal alloy, which may be desirable for heat conducting properties. In some examples, the metal alloy of core **15** may include, but is not limited to, aluminum, Inconel, stainless steel, titanium, copper, nickel, or mixtures and combination thereof. In some examples, core **15** of heat exchanger **14** may be integrally formed (i.e., monolithic or one-piece), and in some examples, may be formed via additive manufacturing. For example, core **15** of heat exchanger **14** may be formed via direct metal laser sintering (DMLS), however, other additive manufacturing processes may be employed, including, but not limited to micro-pen, selective laser sintering, laser wire, electron beam melting, laser engineered net shaping, direct metal deposition, etc. The core may be formed by other suitable manufacturing techniques such as brazing. In some examples, heat exchanger **14** may have total dimensions of about 2 inches (in.) to about 24 inches (in.) wide by about 2 in. to about 24 in. high by about 2 in. to 72 inches in. long. It should be noted, however, that heat exchanger **14** may have any desired width, length and height for a given space within vehicle **12** or for the particular application. It should also be noted that although the example heat exchangers illustrated and discussed herein are rectangular, the core **15** of the heat exchanger **14** and/or the heat exchanger **14** may be any desired shape (e.g., substantially cylindrical or conformal in nature).

(15) In some examples, heat exchanger **14** may include a housing or manifold (not pictured in FIG. **1**) around core **15**. The housing may include a top housing plate and a bottom housing plate which may generally define the working fluid inlets and the outlets of core **15**. For example, the housing may define hot inlet **30**, cold inlet **32**, first outlet **34**, and second outlet **36**. It should be noted that although described and illustrated herein as a plate fin heat exchanger **14**, heat exchanger **14** may be another type of heat exchanger that includes a first flow channel and a second flow channel which are separated by a fin.

(16) FIG. **2** is a conceptual diagram illustrating an example of a cross-flow, plate fin heat exchanger

core **15** of heat exchanger **14** of FIG. **1**. Core **15** may generally be described similarly with respect to FIG. **2**, with additional detail as described below. Core **15** may include first or hot passages **21** (only a single hot passage is labelled) extending substantially perpendicular to second or cold passages **22** (only a single cold passage is labelled). Hot passage **21** is separated from cold passage **22** by tube sheet **27**. Hot enclosure bar **24** may define (e.g., frame) hot passage **21** by defining opposing lateral sides of hot passage **21**. Hot fins **26** may be positioned between hot enclosure bars **24**. Similarly, cold enclosure bar **25** may define (e.g., frame) cold passage **22** by defining opposing lateral sides of cold passage **22**. Cold fins **28** may be positioned between cold enclosure bar **25**. Corner **29** of core **15** may define a shaped, curvilinear (e.g., defining a full circle) corner **29** or a linear, square corner having a 90-degree angle. Although described and illustrated herein primarily with respect to cross-flow, plate fin heat exchangers, it should be noted that other heat exchanger configurations are considered. For example, core **15** may define, in some examples, a counterflow arrangement, a multi-passed crossflow arrangement, or another arrangement.

(17) Core **15** may include a plurality of tube sheets **27** (only an individual tube sheet is labelled in FIG. **2** for clarity) which separate alternating hot passages **21** and cold passages **22** of heat exchanger core **15**. Each respective cold passage **22** of core **20** includes cold enclosure bar **25** and cold fins **28** (only individual cold enclosure bar **25** and cold fin **28** are labeled for clarity). Each hot passage **21** of core **20** includes hot enclosure bar **24** and hot fins **26** (only individual hot enclosure bar **24** and hot fin **26** are labeled for clarity). Hot fin **26** separates adjacent hot channels **40A** and **40B** ("hot channels **40**") Similarly cold fin **28** separates adjacent hot channels **42A** and **42B** ("cold channels **42**").

(18) FIG. **3A** is a conceptual top view of a portion of an example core **115** and example associated fins **152** looking at one fin passage. Example fins **152** may be examples of hot fins **26** or examples of cold fins **28** of FIG. **2**. FIG. **3A** illustrates fins **152** from a top view, located on top of tube sheet **127**. FIG. **3B** illustrates example fin **154** of plurality of fins **152** from a side view. Reference lines connecting FIGS. **3A** and **3B** are illustrated for understanding of the position of at least one aperture **170** on fin **154**.

(19) Fins **152** define a plurality of flow channels **153** in core **115** of heat exchanger **114**. Flow channels **153** may combine to make up a passage, such as hot passage **21** or cold passage **22** as described above with respect to FIG. **2**. Flow channels **153** guide fluid **172** (illustrated by the thick dashed line in FIG. **3A**) through core **115**, and may assist in distributing the flow of fluid and/or provide additional effective heat transfer surface area to improve the heat transfer efficiency of heat exchanger **114**. Flow channels **152** may extend from inlet manifold **174**, which may be inlet **30** or inlet **32** (FIG. **1**), to outlet manifold **174**, which may be first outlet **34** or second outlet **36** (FIG. **1**). Although described below with respect to fin **154** separating first flow channel **156** from adjacent second flow channel **162** for brevity below, in some examples the description of fin **154** may apply to any portion (e.g., every other fin of fins **152**) or all of fins **152**. Similarly, the description of first flow channel **156** and second flow channel **162** may apply to any portion of flow channels **153** or all of flow channels **153**.

(20) Core **115** defines first flow channel **156**, which is a passageway for fluid **172** (e.g., working fluid **16** or **18**, FIG. **1**) extending from first flow channel inlet **158** to first flow channel outlet **160**. Core **115** further defines second flow channel **162** extending from second flow channel inlet **164** to second flow channel outlet **166**. Fin **154** separates first flow channel **156** from second flow channel **158**. In some examples, fin **154** may be a relatively thin wall of form about 0.001 to about 0.01 inches thick, for example from about 0.003 to about 0.005 inches thick. Fin **154** may extend in the Z-direction from a bottom tube sheet (**27**, FIG. **2**) to a top tube sheet (**27**, FIG. **2**).

(21) In some examples, first flow channel **156** may define first flow channel width $L_{sub.W}$, which may be defined as the distance in the Y-direction between fin **154** and an adjacent fin of fins **153** across first flow channel **156**. In some examples, the distance between fin **154** and an adjacent fin may be in a range of from about 0.040 inches to about 0.400 inches, such as for example about

0.060 to about 0.100 inches. In some examples, the distance between the bottom tube sheet and the top tube sheet may be substantially similar to the distance between fin **154** and an adjacent fin, such that first flow channel **156**, second flow channel **162**, or both may define a substantially square cross-section. Alternatively, a distance between a top tube sheet (**27**, FIG. **2**) and a bottom tube sheet (**27**, FIG. **2**) may be of a larger magnitude or a smaller magnitude than the distance between fin **154** and an adjacent fin, such that first flow channel **156** may be longer in the Z-direction or shorter in the Z-direction than first flow channel **156** is wide in the Y-direction.

(22) FIG. **3A** schematically illustrates core **115** during operation in an extreme environment, as described above. In operation, FOD **176** may constrict first flow channel inlet **158** or second flow channel inlet **164**, such that the flow channel **156**, **162** ingests a reduced or eliminated flow of fluid **172** from inlet manifold **174**. Although second flow channel **162** is illustrated and described as the constricted flow channel, core **115** is configured to operate in reverse if first flow channel **156** becomes constricted while second flow channel **162** remains open. Regardless of which flow channel inlet (first flow channel inlet **158** or second flow channel inlet **164**) becomes constricted by FOD during operation, the corresponding flow channel is not available to do heat transfer and the heat transfer surface area associated with the flow channel is wasted on the side that is blocked. Consequently, the performance of heat exchanger **114** may be reduced.

(23) However, fin **154** advantageously defines at least one aperture **170** configured to allow fluid **172** to flow between first flow channel **156** and second flow channel **162** if one of first flow channel inlet **158** or second flow channel inlet **164** becomes constricted through FOD **176**. In some examples, flow of fluid **172** in second flow channel **162** may be reduced because second flow channel inlet **164** is partially constricted, and second channel **162** may receive fluid **172** from first flow channel **156** to boost the flow of fluid **172** in second channel **162**, which may improve heat transfer in core **115**. In some examples, as illustrated for understanding, it is also considered that FOD **176** may constrict flow of fluid **172** through second flow channel inlet **164** such that no fluid **172** may flow into second flow channel **162**. It is also considered that the flow of fluid **172** may be reduced, but not totally stopped. First flow channel inlet **158** may be relatively less constricted by FOD **176**, and thus may pass a larger amount of fluid **172** through first flow channel **156** to perform heat transfer. In such example, absent at least one aperture **170** defined by fin **154**, second flow channel **162** be an unutilized or underutilized portion of core **115**. However, fluid **172** may flow through at least one aperture **170** from first flow channel **156** to second flow channel **162**, thus mitigating thermal performance losses due to the constriction of second flow channel **164**, because at least some fluid may flow through at least a portion of second flow channel **162** from at least one aperture **170** to second flow channel outlet **166**.

(24) Furthermore, in some examples, communication of fluid **172** between first flow channel **156** and second flow channel **162** through at least one aperture **170** may allow for easier release and/or automatic detachment of FOD **176** from second flow channel inlet **164**. For example, as fluid **172** flows into second flow channel **162** through at least one aperture **170**, the pressure of fluid **172** within second flow channel **162** may increase. Thus, the suction force on FOD **176** which keeps FOD **176** on core **115** at second flow channel inlet **164** may be reduced. As this suction force is reduced, more FOD **176** may detach from the face of core **115** at inlet manifold **174**. Core **115** may therefore be less likely to become constricted by FOD **176** and/or stay constricted by FOD **176** by inclusion of fin **154** which defines at least one aperture **170**.

(25) In some examples, due to the constriction of second flow channel inlet **164** with FOD **176**, a reduced fluid pressure of fluid **172** in second flow channel **162** relative to first channel **156** may drive fluid **172** to pass through at least one aperture **170** from first channel **156** into second channel **162**. Additionally, or alternatively, as will be discussed below with respect to FIGS. **4A-5** a primary driver for the flow of fluid **172** across fin **154** may be inertia due to the momentum of fluid **172** in first flow channel **156**.

(26) Fin **154** may define first end **178** and second end **180**. In some examples, as illustrated in FIG.

3A, fin **154** may extend substantially linearly (e.g., linearly or nearly linearly) in the X-direction from first end **178** to second end **180**, which may be called a “straight or plain fin” architecture. Other architectures are considered where the fin may not define a linear shape in the X-direction, as will be described below with respect to the “wavy fin” architecture. Example “wavy fin” architectures will be described below with respect to FIGS. 4A-5. Fin **154** defines fin length $L_{sub.F}$, measured along the X-axis from first end **178** to second end **180**.

(27) In some examples, at least one aperture **170** may be a plurality of apertures defined along the entire fin length. However, at least one aperture **170** may be positioned advantageously to maximize the efficiency of heat exchanger **114** while mitigating losses due to FOD **176** constriction. For example, fin **154** may define solid fin portion **SP1**, which is a portion of fin length $L_{sub.F}$ of fin **154** that does not define any apertures through fin **154**. Solid fin portion **SP1** may be located at a downstream end, relative to the flow of fluid **172** from inlet manifold **174** to outlet manifold **176**, of fin **154**. Solid fin portion **SP1** may define length $L_{sub.SP1}$, which may extend, for example, from about the midpoint along fin length $L_{sub.F}$ to second end **180**. As such, after fluid **172** enters second flow channel **162**, fin **154** may be solid, which may advantageously increase heat transfer area of fin **154**, minimize swirling of fluid **172**, or the like. In some examples, the ratio of the length of solid fin portion $L_{sub.SP1}:L_{sub.F}$ may be in a range from about 0.5:1 to about 0.9:1, such as about 0.8:1 to about 0.9:1. In some examples, Solid fin portion **SP1** may be a first solid fin portion of a plurality of solid fin portions disposed along fin length $L_{sub.F}$ of fin **154**.

(28) Fin **154** may further define porous fin portion **PP**, which is a portion of fin length $L_{sub.F}$ of fin **154** that defines one or more apertures **170** through fin **154**. In some examples, as illustrated, the one or more apertures **170** in porous fin portion **PP** may be a single aperture. Alternatively, porous fin portion **PP** may define a plurality of apertures, such as two, three, four, or more than four apertures. Porous fin portion **PP** may define a length $L_{sub.PP}$ along fin length $L_{sub.F}$ between first end of porous fin portion **182**, which is closer to first end **178** of fin **154**, and a second end of porous fin portion **184**, which is the end of porous fin portion **PP** that is closer to second end **180** of fin **154**. In some examples, the ratio of length $L_{sub.PP}:L_{sub.F}$ may, in some examples, be in a range of from about 0.01:1.0 to about 0.5:1.0.

(29) Fin **154** also defines a fin height $H_{sub.F}$ in the Z-direction. In some examples, in addition to or in the alternative to being defined along length $L_{sub.F}$ of fin **154**, porous fin portion **PP** may be defined along the height of fin **154**. For example, a ratio of $H_{sub.PP}:H_{sub.F}$ may be between about 0.7:1.0 and about 0.9:1.0. Sizing porous portion along height $H_{sub.F}$ of fin **154** in this way may feed fluid **172** into blocked passage **172** while maintaining the required structural integrity of fin **154**.

(30) In some examples, porous fin portion **PP** may be positioned advantageously along fin length $L_{sub.F}$. For example, first end of porous fin portion **182** may be positioned closer to first end **178** of fin **154** than second end **180** of fin **154**, such as between about 0% and about 20% of $L_{sub.F}$ from first end **178** of fin **154**. In some examples, the distance between first end **178** of fin **154** and first end **182** of porous fin portion **PP** may be between about 0.25 inches and about 4.0 inches, such as for example about 0.5 inches and about 1.0 inch. As such, fin **154** may define second solid fin portion **SP2**, which has length $L_{sub.SP2}$ defined between first edge **178** of fin **154** and first edge **182** of porous fin portion **PP**. Positioning porous fin portion **PP** along fin **154** in this way may advantageously allow for a larger portion of the length of second flow channel **162** to receive fluid **172** and be used for heat transfer during operation, and inclusion of second solid fin portion **SP2** may position at least one aperture **170** remotely enough from flow channel inlets **158**, **164** to mitigate constriction of at least one aperture **170**. Advantageously, inclusion of at least one aperture **170** defined by fin **154** into core **115** may open core **115**, resulting in a lower pressure drop across core **115** as more of core **115** is available for flow of fluid **172**.

(31) In some examples, as mentioned above, first flow channel **156** defines flow channel width

L.sub.W. Flow channel width L.sub.W may be of any suitable magnitude, and may be selected according to the heat transfer requirements, pressure drop requirements, and/or fin density of heat exchanger **114**. As discussed above, at least one aperture **170** of porous fin portion PP may be a single aperture **170** which defines aperture width D. Aperture width D may be defined as the greatest distance between any opposing edges of aperture **170**. In some examples, aperture width D may have a greater magnitude than first flow channel width L.sub.W. Arranging core **115** in this way may beneficially mitigate the likelihood that at least one aperture **170** becomes constricted during operation of heat exchanger **114**, because any particle of FOD **176** (e.g., a pebble) that enters first flow channel **156** through first flow channel inlet **158** may subsequently pass-through aperture **170** without constricting or blocking aperture **170** since aperture **170** may be large enough to accommodate such a particle. In some examples, aperture width D may be between about 0.01 inches to about 0.2 inches to about 0.02 inches to about 0.1 inches. Although illustrated and primarily described herein as substantially circular, it should be noted that at least one aperture **170** may have any suitable geometric shape. Some non-limiting examples for the shape of at least one aperture include quadrilateral, hexagonal, elliptical, hemispherical, oval-shaped, or the like. In some examples, aperture width D allow FOD **176** particles with a diameter of 0.040 inches to pass through core **114**.

(32) At least one aperture **170** defined in fin **154** may, in some examples, provide for efficiency mitigation of losses due to FOD contamination. A combination of at least one aperture defined in fin **154** may be advantageously used in combination with other FOD mitigation techniques to capture further benefits. Some complementary FOD mitigation techniques may include, but are not limited to, FOD shields, FOD filters, FOD screens, FOD Fins and combinations thereof, or the like. However, unlike many other solutions which add components and thus increase the weight and volume of heat exchanger **114** to manage or mitigate FOD, the inclusion of at least one aperture **170** in fin **154** may advantageously reduce the weight of heat exchanger **114** by removing relatively dense metallic material from core **115** in select areas designed to improve the efficiency of heat exchanger **114** in extreme environments.

(33) FIGS. **4A** and **4B** illustrate example core **215** of heat exchanger **214**. Core **215** may be an example of core **15** from FIG. **1**. Core **215** of FIGS. **4A-4B** may be generally described similarly to core **115** of FIGS. **3A-3B**, differing as described below, where like reference characters illustrate like elements.

(34) FIG. **4A** is a conceptual top view of a portion of an example core **215** and example associated fins **252**. Example fins **252** may be examples of hot fins **26** or example cold fins **28** of FIG. **2**. FIG. **4A** illustrates fins **252** from a top view, located on top of tube sheet **227**. FIG. **4B** illustrates example fin **254** of plurality of fins **252** from a side view. FIG. **5** illustrates example fin **354** of plurality of fins **252** from a side view. Reference lines connecting FIGS. **4A** to FIGS. **4B** and **5** are illustrated for understanding of the position of porous fin portions on fin **154** relative to the peaks and valleys defined by fin **254**.

(35) In this example, core **215** of heat exchanger **214** has a “wavy fin” architecture. As such, wavy fin **254** defines a series of peaks **286A**, **286B**, **286C** (collectively “peaks **286**”). Peak **286A** may be defined as a point defining a local extremity of wavy fin **254** toward the top of the page in the Y-direction. Wavy fin **254** may also define a series of valleys **288A**, **288B** (collectively “valleys **288**”). Valley **288A** may be defined as a point of local extremity of wavy fin **254** toward the bottom of the page in the Y-direction opposite peaks **286**. As such, wavy fin **254** may not be linear along the X-axis, rather varying between peaks **286** and valleys **288** and wavy fin **254** extends from inlet manifold **274** and outlet manifold **276**.

(36) Wavy fin **254** is configured to change the direction of flow of fluid (**172**, FIG. **3A**, not illustrated in FIG. **4A**) through first flow channel **256** and/or second channel **262** one or more times as fluid **272** flows through core **215**, which may beneficially improve heat transfer within core **215**. Without wishing to be bound by theory, wavy fin **254** may improve heat transfer by encouraging

interaction (e.g., contact) of fluid (172, FIG. 3A) with heat transfer surfaces (e.g., fins 252, tube sheet 227) and/or increasing the residence time of fluid (172, FIG. 3A) within core 215 such that further heat transfer can take place over a longer distance traveled and/or over a longer duration. (37) In some examples, wavy fin 254 may define any suitable shape as it extends between inlet manifold 274 and outlet manifold 276, defining peaks 286 and valleys 288. For example, wavy fin may include linear segments disposed angularly to each other, or define a substantially sinusoidal shape, a curvilinear shape, or combination of shapes as it extends between inlet manifold 214 and outlet manifold 276. In some examples, wavy fin 254 may define relatively straight lengths 290A, 290B (collectively, “relatively straight lengths 290”), which may be disposed between adjacent peaks 286 and valleys 288. It should be noted that although illustrated and discussed herein as relatively straight lengths 290, relatively straight lengths 290 of fin 254 need not be linear, and may only be relatively less inducing of a change of direction of fluid (172, FIG. 3A) than peaks 286 and valleys 288.

(38) In some examples, core 215 may be configured such that relatively straight length 290A is disposed between peak 286A and valley 288A, relatively straight length 290 is disposed between valley 288A and peak 286B, and so on. Accordingly, in operation, fluid (172, FIG. 3A) may flow within first flow channel 256 along straight length 290A, change direction at valley 288A, and subsequently flow within first channel 256 along relatively straight length 290B until changing direction at peak 286B.

(39) With concurrent reference to FIGS. 4A and 4B, in some examples, wavy fin 254 may include one or more porous fin portions PP1, PP2, PP3, PP4, PP5 (collectively “porous fin portions PP”). Each porous fin portion may define one or more than one aperture. For example, porous fin portion PP1 may define aperture 270A, porous fin portion PP2 may define aperture 270B, porous fin portion PP3 may define aperture 270C, porous fin portion PP4 may define aperture 270D, and porous fin portion PP5 may define aperture 270E. In like manner, solid portions SP1-SP6 may be defined along the length of wavy fin 254 between respective porous fin portions PP.

(40) In some examples, as illustrated, one or more of porous fin portion PP, such as porous fin portion PP1, may be defined along a length of wavy fin 254 near a peak 286A of peaks 286 or a valley 288A of valleys 288. Inclusion of a porous fin portion PP1 which defines one or more apertures 170 near a peak 286A or a valley 288A may promote (e.g., improve) cross-channel communication of fluid (172, FIG. 3A) flowing through core 215, which may be desirable if first flow channel inlet 258 or second flow channel inlet 264 becomes constricted by FOD (176, FIG. 3A). Advantageously, the wavy fin arrangement of core 215 may be beneficial even if FOD (176, FIG. 3A) is not present, as the geometry of wavy fins 253 defining porous fin portion(s) PP may effectively introduce flow turbulence into those fin channels that are exhibiting semi-stagnate flow. For example, in addition to the reduced pressure in second channel 262 (in the example with constricted second flow channel inlet 264) causing fluid (172, FIG. 3A) to flow into second flow channel 262, the momentum of fluid (172, FIG. 3A) flowing in first flow channel 256 may drive fluid (172, FIG. 3A) through one or more apertures 270 in porous fin portion PP1 when porous fin portion PP1 is positioned near peak 286A or valley 288A. As discussed herein, porous fin portion PP1 may be considered to be near valley 288A when it is positioned between the midpoints of relatively straight portions 290A and 290B. In some examples, porous fin portion PP1 may be positioned along a length of wavy fin 254 that defines valley 288A. Furthermore, and advantageously, the wavy fin arrangement of core 215 may be beneficial even if FOD (176, FIG. 3A) is not present, as the geometry of wavy fins 252 defining porous fin portion(s) PP may effectively introduce flow turbulence into those flow channels 253 that may be exhibiting semi-stagnate flow.

(41) In some examples, as illustrated, wavy fin 254 may define porous fin portions PP, and each porous fin portion PP may correspond to each peak 286A of peaks 286 and each valley 288A of valleys 288. Cross-channel communication may be even more improved in such examples. This

effective cross communication will promote fluid (172, FIG. 3A) mixing and may improve heat transfer in core 215.

(42) FIG. 5 illustrates example wavy fin 354 from a side view. Wavy fin 354 may be generally described as wavy fin 254 of FIG. 4B, differing as described below, where like reference characters indicate like elements.

(43) In some examples, as mentioned above, porous fin portion PP1 may define more than one aperture, such as aperture 371, aperture 373, and aperture 375, (collectively, “apertures 370”). Apertures 370 may include any suitable number of apertures and may be arranged in any suitable sizes or shapes. For examples, apertures 370 may be small perforations in wavy fin 354 which may distribute fluid (172, FIG. 3A) as it crosses wavy fin 354 into the adjacent flow channel. Apertures 370 may be arranged in an orderly fashion, such as in rows and/or columns across the face of wavy fin 354, or may be distributed randomly.

(44) Wavy fin 354 defines first end 378, second end 380, peak 386A, and valley 388B. It may be desirable, in some examples, to shift the location of porous fin portion PP1 downstream along arrow A (i.e. toward second end 380 of wavy fin 354) such that the center of porous portion PP1 is downstream of peak 386A. In some examples, as illustrated, porous fin portion PP1 may still be positioned near peak 386A along the length of fin 354. Positioning porous fin portion PP1 in this way may improve cross channel communication in core 315 by allowing a vector defined by the momentum of fluid (172, FIG. 3A) flowing in the second flow channel (262, FIG. 4A) to intersect with wavy fin 354 at a location at or near the center of porous fin portion PP1. Additionally, or alternatively, in some examples, porous fin portion PP1 may be shifted in a direction opposite arrow A such that PP1 may be defined in a location along wavy fin 354 configured such that fluid (172, FIG. 3A) flows past porous fin portion PP1 before peak 386A. Stated similarly, in some examples porous fin portion PP1 may be positioned upstream of peak 386A.

(45) In some examples, wavy fin 354 may defines a fin surface area, which may be the area occupied by the fin along the X-Z plane. In some examples, a portion of the fin surface area defines apertures 370. In some examples, the area which defines apertures 370 may be between about 0.1% and about 20% of the fin surface area, such as between about 0.5% and about 5% of the fin surface area.

(46) FIG. 6 is a flowchart illustrating an example technique for manufacturing a heat exchanger according to the present disclosure. Although described primarily with respect to heat exchanger 114 of FIG. 3A and heat exchanger 214 of FIG. 4A, techniques according to the present disclosure may be suitable for forming other heat exchangers, such as heat exchanger 14 of FIG. 1, and the described heat exchangers may be formed using other techniques.

(47) With concurrent reference to FIGS. 3A and 3B, the technique of FIG. 6 includes forming first flow channel 156 of heat exchanger 114 (400). First flow channel 156 includes first flow channel inlet 158 and first flow channel outlet 160.

(48) The technique of FIG. 6 also includes forming second flow channel 162 of heat exchanger 114 (402). Second flow channel 162 includes second flow channel inlet 164 and second flow channel outlet 166. The technique of FIG. 6 also includes separating first flow channel 156 from second flow channel 162 with fin 154 which defines at least one aperture 170 (404). At least one aperture 170 allows fluid 172 to flow between first flow channel 156 and second flow channel 162 if one of the first flow channel inlet 158 or second flow channel inlet 164 becomes constricted through a buildup of FOD 176 or other contaminants.

(49) In some examples, fin 154 defines fin length L.sub.F, and in some examples a portion of the fin length L.sub.F defines solid fin portion SP1 that does not define any apertures, and another portion of fin length L.sub.F defines a porous fin portion PP that includes at least one aperture 170. Turning to FIGS. 4A, 4B, and FIG. 6, in some examples wavy fin 254 defines a series of peaks 286 and valleys 288 along fin 254. Peaks 286 and valleys 288 may be configured to change the direction of flow of fluid (172, FIG. 3A) through first flow channel 256 and/or the second flow

channel **262**. In some examples, solid fin portion **SP1** may be disposed along a relatively straight length **290A** of wavy fin **254**. In some examples, porous fin portion **PP1** may be disposed along length of wavy fin **254** near peak **286A** or valley **288A**. In some examples, wavy fin **254** may define a plurality of porous fin portion **PP**, and each respective porous fin portion (**PP1**, **PP2**, **PP3**, **PP4**, **PP5**) may correspond to each respective peak and valley defined by wavy fin **254** (**286A**, **288A**, **286B**, **288B**, **286C**).

(50) Referring back to FIG. **3B**, in some examples, at least one aperture defined by porous fin portion **PP** may be a single aperture **170**, and single aperture **170** may be defined by fin **154** in porous portion **PP** of fin **154** which is nearer to fluid inlet **174** than fluid outlet **176**. In some examples, first flow channel **156** may define flow channel width $L_{sub.W}$, the single aperture **170** may define single aperture width D , and single aperture width D may have greater magnitude than first flow channel width $L_{sub.W}$. In some examples, the technique of FIG. **6** may further include manufacturing vehicle **12** (FIG. **1**) with heat exchanger **14** as a component. In some examples, vehicle **12** may be an aircraft such as a helicopter.

(51) The following numbered clauses illustrate one or more aspects of the devices and techniques described in this disclosure.

(52) Clause 1: A heat exchanger comprising: a first flow channel having a first flow channel inlet and a first flow channel outlet; a second flow channel having a second flow channel inlet and a second flow channel outlet; and a fin separating the first flow channel from the second flow channel, wherein the fin defines at least one aperture configured to allow fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of foreign object debris.

(53) Clause 2: The heat exchanger of clause 1, wherein the fin is a plain fin which extends substantially linearly from a first end to a second end.

(54) Clause 3: The heat exchanger of clause 1 of clause 2, wherein the fin defines a fin length, wherein a portion of the fin length defines a solid fin portion that does not define any apertures, and wherein a portion of the fin length defines a porous fin portion including the at least one aperture.

(55) Clause 4: The heat exchanger of any of clauses 1-3, wherein a first end of the porous fin portion is between about 0% and about 20% of the fin length away from a first end of the fin, and the porous fin portion is configured to reduce a pressure drop across a core of the heat exchanger.

(56) Clause 5: The heat exchanger of clause 3, wherein the fin is a wavy fin defining a series of peaks and valleys along the fin length, the peaks and valleys configured to change the direction of flow through the first flow channel or the second flow channel, wherein the solid fin portion is disposed along a relatively straight length of the fin, and wherein the porous fin portion is disposed along length of the fin near a peak or a valley.

(57) Clause 6: The heat exchanger of clause 5, wherein the fin defines a porous fin portion that corresponds to each peak and a porous fin portion that corresponds to each valley defined by the fin.

(58) Clause 7: The heat exchanger of clause 3, wherein the solid fin portion is a first solid fin portion, the porous fin portion is a first porous fin portion, and the fin further defines a second solid fin portion and a second porous fin portion along the fin length.

(59) Clause 8: The heat exchanger of any of clauses 1-7, wherein the at least one aperture is a single aperture, and the single aperture is defined by the fin in a portion of the fin that is nearer to the fluid inlet than the fluid outlet.

(60) Clause 9: The heat exchanger of clause 8, wherein the first flow channel defines a flow channel width, the single aperture defines a single aperture width, and the single aperture width has greater magnitude than the first flow channel width.

(61) Clause 10: The heat exchanger of any of clauses 1-9, wherein the heat exchanger is a component of an aircraft.

(62) Clause 11: A method, the method comprising: forming a first flow channel of a heat

exchanger, the first flow channel having a first flow channel inlet and a first flow channel outlet; forming a second flow channel of the heat exchanger, the second flow channel having a second flow channel inlet and a second flow channel outlet; and separating the first flow channel from the second flow channel with a fin, wherein the fin defines at least one aperture which allows the fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of foreign object debris.

(63) Clause 12: The method of clause 11, wherein the fin is a plain fin which extends substantially linearly from a first end to a second end.

(64) Clause 13: The method of clause 11 or clause 12, wherein the fin defines a fin length, wherein a portion of the fin length defines a solid fin portion that does not define any apertures, and wherein a portion of the fin length defines a porous fin portion including the at least one aperture.

(65) Clause 14: The method of clause 13, wherein a first end of the porous fin portion is between about 0% and about 20% of the fin length away from a first end of the fin, and the porous fin portion is configured to reduce a pressure drop across a core of the heat exchanger.

(66) Clause 15: The method of clause 13, wherein the fin is a wavy fin defining a series of peaks and valleys along the fin length, the peaks and valleys configured to change the direction of flow through the first flow channel or the second flow channel, wherein the solid fin portion is disposed along a relatively straight length of the fin, and wherein the porous fin portion is disposed along length of the fin near a peak or a valley.

(67) Clause 16: The method of clause 15, wherein the fin defines a porous fin portion that corresponds to each peak and a porous fin portion that corresponds to each valley defined by the fin.

(68) Clause 17: The method of clause 13, wherein the solid fin portion is a first solid fin portion, the porous fin portion is a first porous fin portion, and the fin further defines a second solid fin portion and a second porous fin portion along the fin length.

(69) Clause 18: The method of any of clauses 11-17, wherein the at least one aperture is a single aperture, and the single aperture is defined by the fin in a portion of the fin that is nearer to the fluid inlet than the fluid outlet.

(70) Clause 19: The method of clause 18, wherein the first flow channel defines a flow channel width, the single aperture defines a single aperture width, and the single aperture width has greater magnitude than the first flow channel width.

(71) Clause 20: The method of any of clauses 11-19, wherein the heat exchanger is a component of an aircraft.

(72) Various examples have been described. These and other examples are within the scope of the following claims.

Claims

1. A heat exchanger comprising: a first flow channel having a first flow channel inlet and a first flow channel outlet; a second flow channel having a second flow channel inlet and a second flow channel outlet; and a fin separating the first flow channel from the second flow channel, wherein the fin is a wavy fin defining a series of peaks and valleys along a length of the fin, the peaks and valleys configured to change a direction of flow through the first flow channel or the second flow channel, wherein: a first portion of the length of the fin defines a first solid fin portion that does not define any apertures, and a second portion of the length of the fin defines a first porous fin portion defining a plurality of randomly distributed apertures, the first porous fin portion is configured to allow fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of foreign object debris, wherein the first porous fin portion is located such that a vector defined by a momentum of fluid flowing in the first flow channel or the second flow channel intersects with the

wavy fin at a location at a center of the first porous fin portion, the first porous fin portion extends downstream along a length of the wavy fin from a first end of the first porous fin portion to a second end of the first porous fin portion, wherein the center of the first porous fin portion is downstream of a local peak or valley of the wavy fin, the first end of the first porous fin portion is aligned with a peak or a valley of the wavy fin, the first solid fin portion is disposed along a relatively straight length of the fin, and a second porous fin portion is displaced from the first porous fin portion by a second solid fin portion.

2. The heat exchanger of claim 1, wherein the first end of the first porous fin portion is between 0% and 20% of the length of the fin away from a first end of the fin, and the first porous fin portion is configured to reduce a pressure drop across a core of the heat exchanger.

3. The heat exchanger of claim 1, wherein the fin defines a porous fin portion that corresponds to each peak and a porous fin portion that corresponds to each valley defined by the fin.

4. The heat exchanger of claim 1, wherein the heat exchanger is a component of an aircraft.

5. A method, the method comprising: forming a first flow channel of a heat exchanger, the first flow channel having a first flow channel inlet and a first flow channel outlet; forming a second flow channel of the heat exchanger, the second flow channel having a second flow channel inlet and a second flow channel outlet; and separating the first flow channel from the second flow channel with a fin, wherein the fin is a wavy fin defining a series of peaks and valleys along a length of the fin, the peaks and valleys configured to change the direction of flow through the first flow channel or the second flow channel, wherein: a first portion of the length of the fin defines a first solid fin portion that does not define any apertures, a second portion of the length of the fin defines a first porous fin portion defining a plurality of randomly distributed apertures, the first porous fin portion allows a fluid to flow between the first flow channel and the second flow channel if one of the first flow channel inlet or second flow channel inlet becomes constricted through a buildup of foreign object debris, the first porous fin portion is located such that a vector defined by a momentum of fluid flowing in the first flow channel or the second flow channel intersects with the wavy fin at a center of the first porous fin portion, the first porous fin portion extends downstream along a length of the wavy fin from a first end of the first porous fin portion to a second end of the first porous fin portion, wherein the center of the first porous fin portion is downstream of a local peak or valley of the wavy fin, the first end of the first porous fin portion is aligned with a peak or a valley of the wavy fin, the first solid fin portion is disposed along a relatively straight length of the fin, and a second porous fin portion is displaced from the first porous fin portion by a second solid fin portion.

6. The method of claim 5, wherein the first end of the first porous fin portion is between 0% and 20% of the length of the fin away from a first end of the fin, and the first porous fin portion is configured to reduce a pressure drop across a core of the heat exchanger.

7. The method of claim 5, wherein the fin defines a porous fin portion that corresponds to each peak and a porous fin portion that corresponds to each valley defined by the fin.

8. The method of claim 5, wherein the heat exchanger is a component of an aircraft.

9. The method of claim 5, wherein the plurality of randomly distributed apertures define between 0.5% and 5% of a total surface area of the wavy fin.
